

FTL Syria AI4Climate

Python for Climate Hackathon

Hackathon Dates: 28 September – 2 October 2026

Theme: Using Python and Data for Climate Action

1. Hackathon Challenge

Each team must use **Python and climate-related data** to investigate and respond to a real climate or environmental challenge.

Your project should follow this process:

Climate Problem → Data → Python Analysis → Visualisation → Findings → Proposed Solution / Impact

2. Challenge Areas

Teams may work on one of the following areas:

- Climate and weather
- Water and drought
- Agriculture and food systems
- Renewable energy and energy efficiency
- Air quality and environmental monitoring
- Sustainable cities
- Climate risks and natural hazards

Teams may propose another climate-related topic.

3. Problem Statement – First Deliverable

Each team must submit its **Problem Statement by: Tuesday, 29 September 2026 – 21:00**

Submit:

Team Number:

Project Title:

Climate Challenge:

Problem Statement: Maximum 150 words

Main Research Question: One clear question

Dataset: Name + source/link

Example

Project: Syria HeatWatch

Problem: Increasing temperatures may expose communities to more frequent extreme-heat conditions.

Research Question:

Have extreme-temperature days increased over the last 20 years?

Expected Output:

Python analysis and visualisation showing temperature trends and identifying unusually hot periods.

The problem statement should be **specific, measurable and possible to investigate with available data**.

4. Dataset

Teams can either:

Option A — Recommended

Use data from one of the suggested official sources below.

Option B — External Dataset

Find your own dataset.

External datasets are allowed but must:

- come from a credible source;
- be relevant to the selected problem;
- be accessible to the whole team;
- be clearly referenced;
- contain enough information for meaningful Python analysis.

Recommended Data Sources

World Bank Climate Change Knowledge Portal

Useful for temperature, precipitation, historical climate and climate projections. The portal provides downloadable country and subnational climate data. [Climate Knowledge Portal](#)
[World Bank Climate Data Download](#)

NASA POWER

Useful for temperature, rainfall, solar radiation, wind and other meteorological variables.

Data can be downloaded as CSV and other formats. [POWER NASA POWER Data Portal](#)

FAOSTAT

Useful for agriculture, crop production, land use and food-system analysis. FAOSTAT provides statistics covering more than 245 countries and territories. [FAOHome](#)
[FAO Statistics / FAOSTAT](#)

UN Data

Can be used for additional country-level socioeconomic or environmental indicators.
[UN Data – Syrian Arab Republic](#)

Teams are **not required to use Syria-only data** if an appropriate regional or comparable dataset better supports the selected problem.

5. Minimum Technical Requirements

Each project must include:

Python

- Python code that runs successfully;
- appropriate use of variables, conditions, loops and/or functions;
- clear and readable code.

Data

- import the dataset;
- inspect the data;
- clean or prepare it where necessary;
- explain relevant variables.

Analysis

At least **3 meaningful analyses**, for example:

- averages;
- minimum/maximum;
- changes over time;
- percentage change;
- comparison between periods;
- threshold detection;
- correlation;
- grouping or ranking.

Visualisation

Python must be used as a meaningful part of the solution, especially for data processing, analysis, automation, machine learning or backend logic. Teams may use other programming languages, frameworks and platforms to build the final product, interface or deployment layer.

Solution	Example
Interactive Dashboard	Climate-risk dashboard showing temperature, drought and rainfall indicators
Website	Public climate-information portal for communities
Mobile App	Heatwave or drought alert application
Early-Warning System	Python predicts risk and sends an alert
AI Model	Predict crop risk, rainfall category or energy demand
GIS Application	Interactive map identifying vulnerable locations
Decision-Support Tool	User enters conditions and receives climate recommendations
Monitoring Platform	Dashboard for air quality, water or temperature sensors
API Service	Python backend serving climate predictions to another application
Chatbot / AI Assistant	Climate-information assistant connected to analysed datasets
IoT Concept	Sensors + Python + dashboard for environmental monitoring

6. Hackathon Schedule

Date	Main Activity	Expected Output
28 Sep	Challenge Definition	Select topic and start problem definition
29 Sep	Dataset & Problem Definition	Problem Statement due by 21:00
30 Sep	Python Development	Data cleaning, analysis and initial visualisations
1 Oct	Testing & Improvement	Working project + interpretation of results
2 Oct	Finalisation & Presentation	Final submission

7. Final Submission

Deadline: Friday, 2 October 2026 – 17:00

Each team must submit:

1. **Python Notebook (.ipynb)**
2. **Dataset or dataset link**
3. **Final presentation – maximum 7 slides**
4. **Short project summary**
5. **Colab/GitHub link**, where applicable

8. Final Presentation

Maximum 7 slides:

1. **Project Title & Team**
2. **Climate Problem & Research Question**
3. **Dataset**
4. **Python Approach**
5. **Main Results & Visualisations**
6. **Climate Impact / Proposed Solution**
7. **Limitations & Next Steps**

Presentation Time

5 minutes presentation + 3 minutes Q&A